Xilong Zhou

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Education

Ph.D., Computer Science, Texas A&M University. *Aug 2018 – Aug 2024*
Advisor: Dr. Ergun Akleman and Dr. Nima Kalantari,
Dissertation: Towards Practical and Robust Material Acquisition and Generation.

M.S., Petroleum Engineering, Texas A&M University. *Aug 2014 – Aug 2016*
Advisor: Dr. Jenn-Tai Liang,
Thesis: Bulk and Adsorption Properties of Surfactant Entrapped in Polyelectrolyte Complex Nanoparticles.

B.E., Petroleum Engineering, China University of Petroleum (Beijing). *Sep 2010 – Jun 2014*

Research Interests

Generative AI, Video and 3D Content Generation/Editing, Neural Rendering, View Synthesis, Inverse Rendering

Employments

Postdoc Researcher, Max Planck Institute for Informatics. *Sep 2024 – Present*
Advisor: Dr. Christian Theobalt.

Research Intern, Adobe Research. *Feb – Jul 2024*
Advisor: Dr. Miloš Hašan.

Research Intern, Meta Reality Lab. *Sep – Dec 2022*
Advisor: Dr. Chakravarty R. Alla Chaitanya.

Research Intern, Adobe Research. *May – Aug 2022*
Advisor: Dr. Miloš Hašan.

Research Intern, Adobe Research. *May – Aug 2021*
Advisor: Dr. Miloš Hašan.

Publications

∗ Equal Contribution.

Preprints

[1] **Xilong Zhou**, Jianchun Chen*, Pramod Rao*, Timo Teufel, Linjie Lyu, Tigran Minasian, Oleksandr Sotnychenko, Xiao-Xiao Long, Marc Habermann, and Christian Theobalt. OLATverse: A Large-scale Real-world Object Dataset with Precise Lighting Control. *arXiv preprint, Nov 2025.*

[2] **Xilong Zhou***, Bao-Huy Nguyen*, Loïc Magne, Vladislav Golyanik, Thomas Leimkühler, and Christian Theobalt. Splat the Net: Radiance Fields with Splattable Neural Primitives. *arXiv preprint, Oct 2025.*

[3] **Xilong Zhou**, Pedro Figueiredo, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Yiwei Hu, and Nima Khademi Kalantari. RealMat: Realistic Materials with Diffusion and Reinforcement Learning. *arXiv preprint, Sep 2025.*

Journal/Conference Articles

[4] Pramod Rao, **Xilong Zhou***, Abhimitra Meka*, Gereon Fox, Mallikarjun B R, Fangneng Zhan, Tim Weyrich, Bernd Bickel, Hanspeter Pfister, Wojciech Matusik, Thabo Beeler, Mohamed Elgharib, Marc Habermann, and Christian Theobalt. 3DPR: Single Image 3D Portrait Relighting with Generative Priors. *SIGGRAPH Asia 2025.*

[5] Avinash Paliwal, **Xilong Zhou**, Andrii Tsarov, and Nima Kalantari. PanoDreamer: Optimization-Based Single Image to 360 3D Scene With Diffusion. *SIGGRAPH Asia 2025.*

[6] Timo Teufel*, Pulkit Gera*, **Xilong Zhou**, Umar Iqbal, Pramod Rao, Jan Kautz, Vladislav Golyanik,

and Christian Theobalt. HumanOLAT: A Large-Scale Dataset for Full-Body Human Relighting and Novel-View Synthesis. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025.*

[7] Avinash Paliwal, **Xilong Zhou**, Wei Ye, Jinhui Xiong, Rakesh Ranjan, and Nima Kalantari. RI3D: Few-Shot Gaussian Splatting With Repair and Inpainting Diffusion Priors. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2025.*

[8] **Xilong Zhou**, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Yannick Hold-Geoffroy, Kalyan Sunkavalli, and Nima Khademi Kalantari. PhotoMat: A Material Generator Learned from Single Flash Photos. *SIGGRAPH 2023.*

[9] **Xilong Zhou**, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Kalyan Sunkavalli, and Nima Khademi Kalantari. A Semi-Procedural Convolutional Material Prior. *Computer Graphics Forum (Eurographics 2023).*

[10] **Xilong Zhou**, Miloš Hašan, Valentin Deschaintre, Paul Guerrero, Kalyan Sunkavalli, and Nima Khademi Kalantari. TileGen: Tileable, Controllable Material Generation and Capture. *SIGGRAPH Asia 2022.*

[11] **Xilong Zhou** and Nima Khademi Kalantari. Look-Ahead Training with Learned Reflectance Loss for Single-Image SVBRDF Estimation. *ACM Transactions on Graphics (SIGGRAPH Asia 2022).*

[12] **Xilong Zhou** and Nima Khademi Kalantari. Adversarial Single-Image SVBRDF Estimation with Hybrid Training. *Computer Graphics Forum (Eurographics 2021).*

[13] **Xilong Zhou**, Jenn-Tai Liang, Corbin D Andersen, Jiajia Cai, and Ying-Ying Lin. Enhanced Adsorption of Anionic Surfactants on Negatively Charged Quartz Sand Grains Treated with Cationic Polyelectrolyte Complex Nanoparticles. *Colloids and Surfaces A: Physicochemical and Engineering Aspects, 553, 397-405, 2018.*

Selected Ongoing Research

Physically Plausible Video Generation and Editing *Project Leader*

- Developing a diffusion-based framework for physically plausible video appearance editing and control
- Enables material editing, object manipulation, and lighting-aware adjustments

Realistic Multi-View 3D Generation under Lighting Control *Project Leader*

- Building a diffusion-based 3D generative model trained on real-world data
- Supports consistent view synthesis and controllable lighting

Selected Awards

Travel Grant, CSCE Department, Texas A&M University. *2023*

National First Prize of National Petroleum Engineering Design Competition. *2013*

Honorable Mention of International Mathematical Contest in Modeling. *2013*

National Second Prize of National Mathematics Modeling Contest. *2012*

Media Coverage

SIGGRAPH Thesis Fast Forward, Ph.D. Dissertation. *2024*
Two Minutes Paper, Paper [8]. *2023*

External Service

Reviewer: SIGGRAPH, SIGGRAPH Asia, Eurographics, Pacific Graphics, ICCV, TOG, IEEE TVCG, IEEE TPAMI, CGF, Computers & Graphics.

Teaching

Co-Instructor, Graphics Seminar, Saarland University. *Summer 2025*

Teaching Assistant, Machine Learning, Texas A&M University. *Fall 2020, Spring 2022*

Teaching Assistant, Analysis of Algorithms, Texas A&M University. *Summer 2019, Fall 2021*

Teaching Assistant, Programming, Texas A&M University. *Spring 2021*

Teaching Assistant, Data Structure and Algorithm, Texas A&M University. *Spring 2019*
Teaching Assistant, Discrete Structure for Computing, Texas A&M University. *Fall 2019, Fall 2018*
Teaching Assistant, Computer for Visualization, Texas A&M University. *Spring/Summer 2017*
Teaching Assistant, Formation Evaluation, Texas A&M University. *Spring 2016*
Teaching Assistant, Unconventional Oil and Gas, Texas A&M University. *Fall 2015*